Project Title: Dropout Risk Detection

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- Batch: A1

- Course: ML Mini Project

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Problem Recap

Phase 1 identified that many students drop out due to low academic performance and lack of engagement.

However, the baseline model had limitations in accuracy due to class imbalance and limited model complexity.

Institutes require a more reliable and accurate system to identify at-risk students early.

In Phase 2, the focus is on improving prediction performance using advanced machine learning models and imbalance handling techniques.

Goal: build a more accurate and robust prediction model to support early intervention.

Dataset (Final)

Dataset Source

- **Name:** *Predict Students' Dropout and Academic Success*
- **Source:** Kaggle / UCI Machine Learning Repository

Number of Samples

- **Rows (instances): 4,424 students**
- **Features: 36 input features + 1 target column**

Feature Description

The dataset includes **student demographic, academic, financial, family, and economic details**.

- **Demographic:** gender, age, nationality
- **Academic:** course, previous grades, semester performance
- **Financial:** fees status, scholarship, debtor
- **Family:** parents' education & occupation
- **Economic:** unemployment, inflation, GDP

Data Pre-processing

Handling Missing Values

- No missing values
- Dataset is already cleaned and ready to use

Encoding Categorical Variables

- No additional encoding required
- All categorical fields are already label encoded as numeric values
- Features can be directly used for ML models

Normalization / Scaling

- Use StandardScaler
- Important for Logistic Regression, SVM, DNN
- Less important for tree-based models

Feature Engineering

Selected Features

- **Personal:** age, gender, marital status, nationality, international
- **Academic background:** previous qualification, application mode/order
- **Semester performance:** enrolled, approved units, grades, evaluations
- **Financial/social:** fees status, debtor, scholarship, course
- **Economic/family:** unemployment, inflation, GDP, parents' education

Justification

- **Semester approved units + grades** → strongest academic predictors
- **Tuition fees, debtor, scholarship** → financial stress indicators
- **Age and family background** → personal support & responsibility factors
- **Economic indicators** → external stress on continuation

Domain Knowledge Used

- Low grades → poor academic progress
- Fee delays → financial hardship
- Scholarship → higher motivation
- Older/married students → more outside responsibilities

Models Implemented

Algorithms Shortlisted

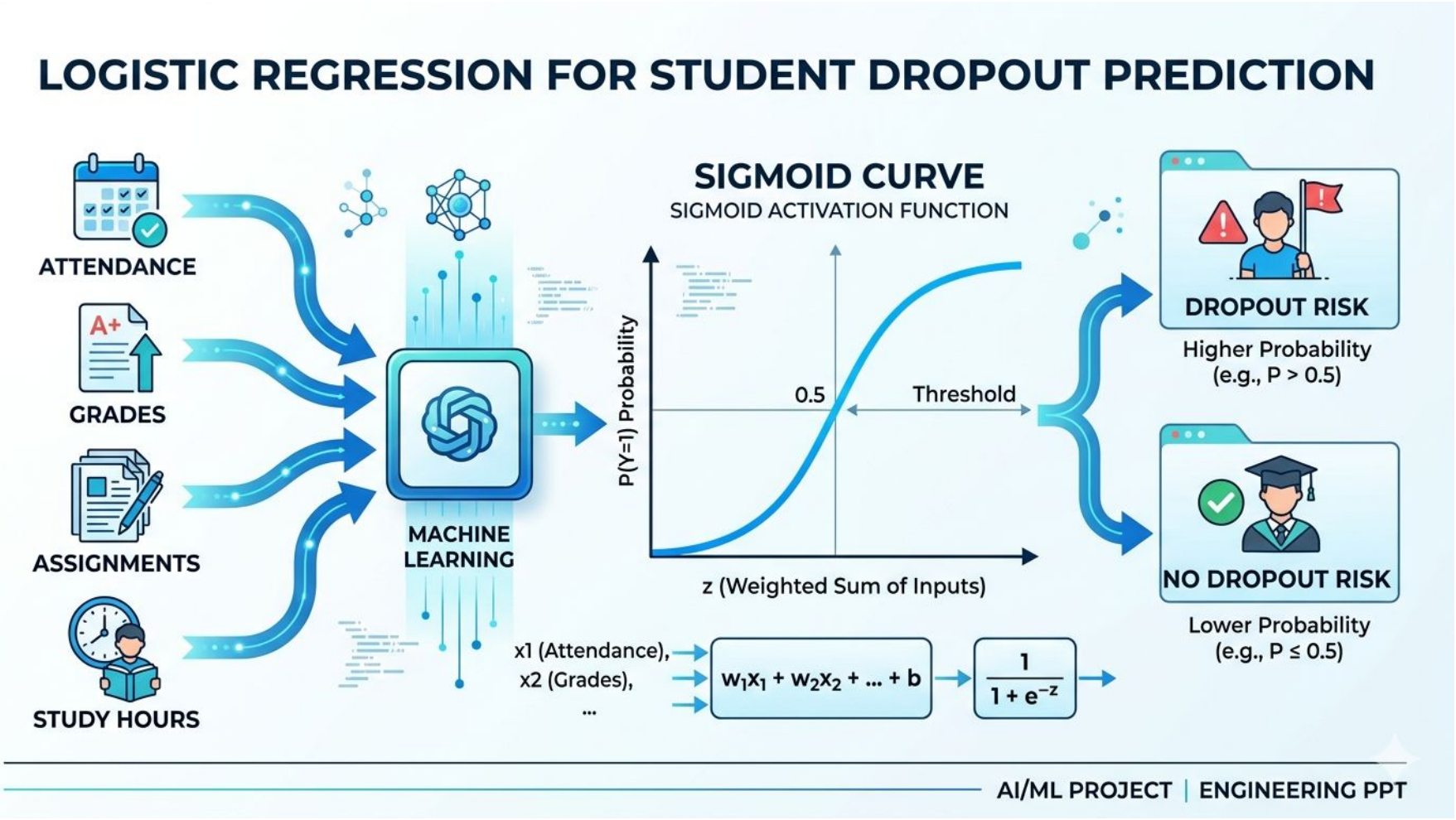
- Logistic Regression
- Random Forest
- Gradient Boosting

Reason for Choosing Each

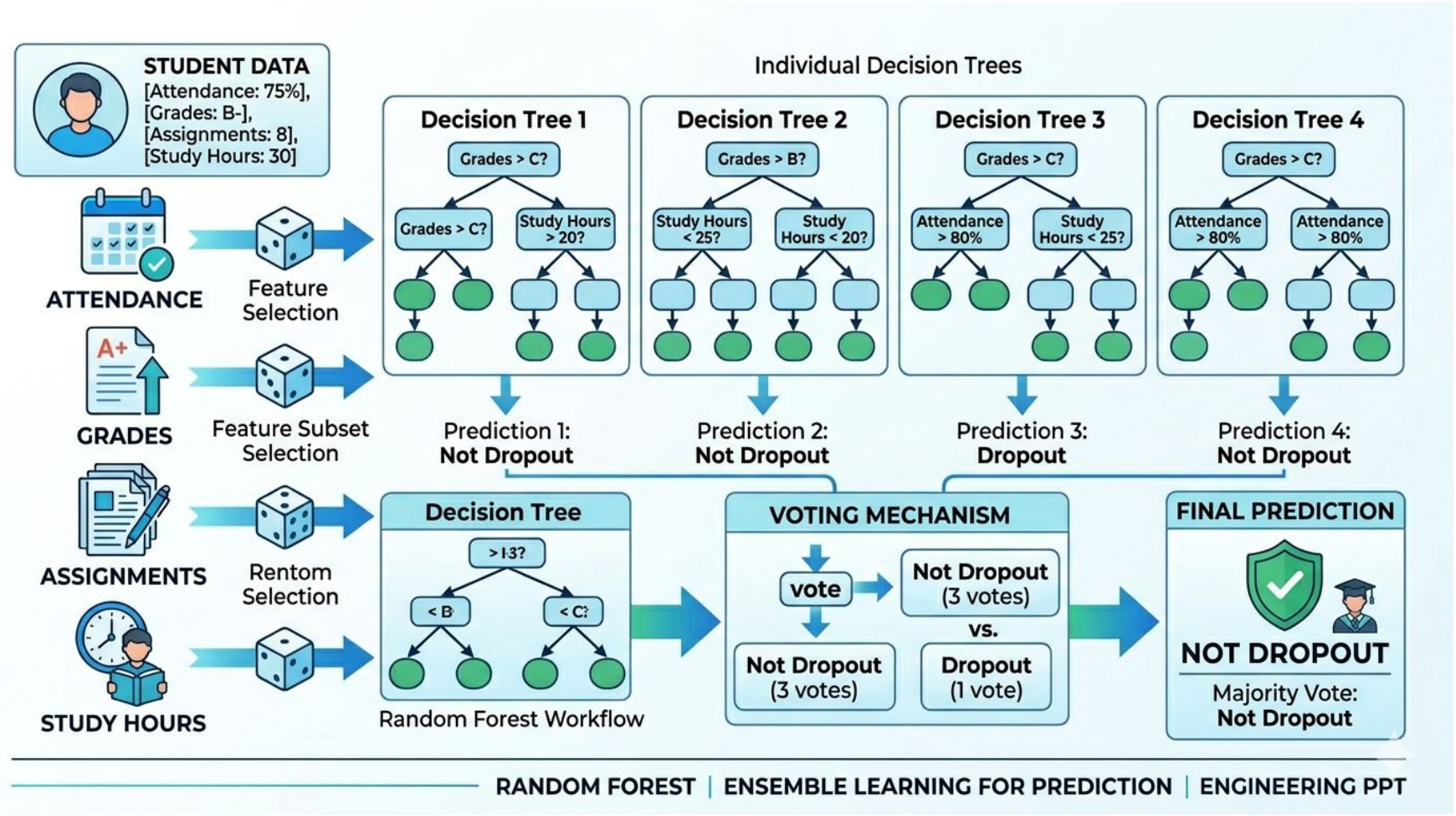
- Logistic Regression: simple, fast, and strong baseline classifier
- Random Forest: handles non-linearity, reduces overfitting, gives feature importance
- Gradient Boosting: high accuracy by learning from previous model errors

Brief description of each (diagram)

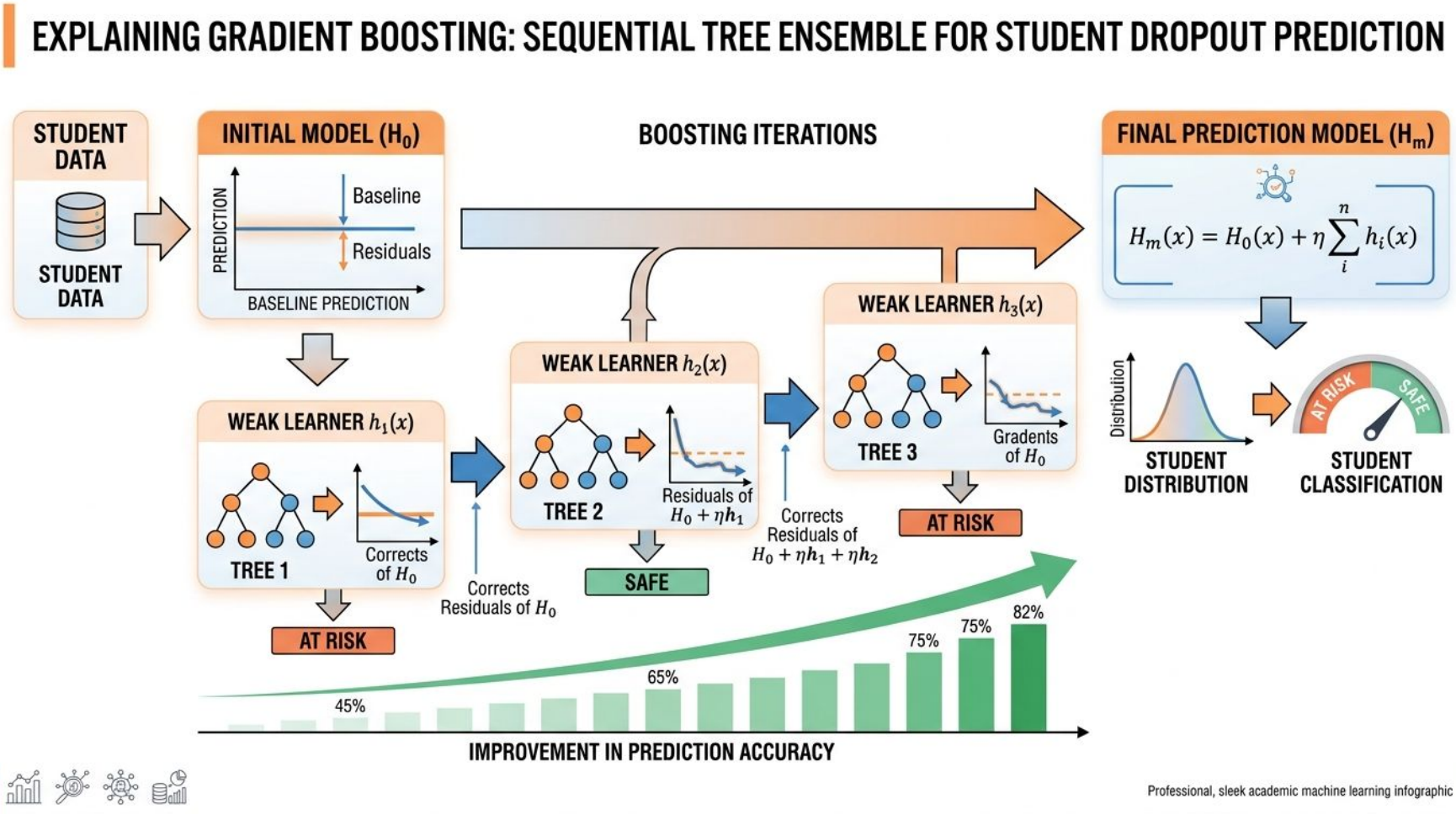
Logistic Regression



Random Forest



Gradient Boosting



Hyperparameter Tuning

Parameters Tuned

For Random Forest, the following parameters were tuned:

- n_estimators → [100, 200]
- max_depth → [None, 10, 20]

For Gradient Boosting, default:

n_estimators = 100

For Logistic Regression:

- max_iter = 1000

Method Used

- GridSearchCV was used for Random Forest
- Manual/default tuning was used for Logistic Regression and Gradient Boosting
- 3-fold cross-validation
- F1-score used as scoring metric

Effect on Performance

- Random Forest performance improved after tuning
- Better balance between precision and recall
- F1-score improved to 81.02%
- Gradient Boosting still achieved the best F1-score (82.14%)
- Helped in selecting the best final model

Evaluation Metrics and Model Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	87.46%	78.36%	84.15%	81.15%
Random Forest	88.25%	84.09%	78.17%	81.02%
Gradient Boosting	88.70%	83.33%	80.99%	82.14%

Best Model Selection

Selected model: Gradient Boosting

Reason for selection:

- It achieved the highest accuracy (88.70%)
- It has the best F1-score (82.14%), which means the best balance between precision and recall
- Precision and recall are both strong, making it reliable for identifying at-risk students
- Performs better than Logistic Regression and Random Forest overall
- Works well on complex patterns and feature interactions
- Suitable for student dropout prediction, where balanced classification is important

Error Analysis

Where model fails:

- Borderline students → students whose behavior is between dropout and non-dropout
- Sudden life events → personal, financial, or health issues not present in the dataset
- Incomplete data → missing attendance, grades, or assignment records
- Noisy data → incorrect or inconsistent student information
- Unseen patterns → new types of student behavior not learned during training
- Class imbalance issues → minority dropout cases may still be harder to predict

Example misclassifications

A student is predicted as “Safe”, but actually drops out.

Example:

- Good attendance and grades initially
- Suddenly faces financial or personal issues
- Leaves college unexpectedly

Model fails because these hidden factors were not in training data.

Improvement over Phase-1

- Previous vs current performance

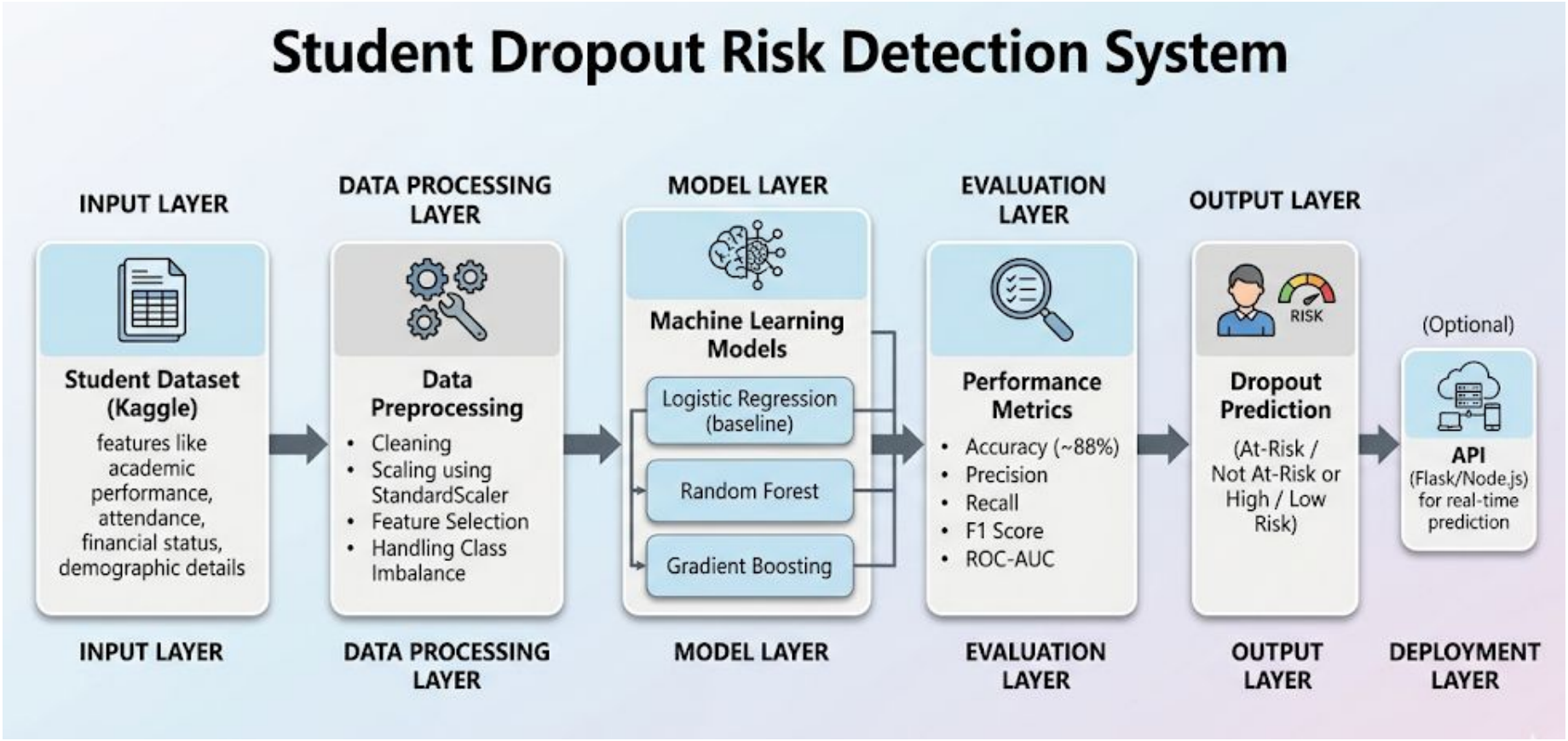
Phase	Model	Accuracy	Precision	Recall	F1 Score
Phase 1	Logistic Regression (Baseline Model)	88.02%	87.39%	73.24%	79.69%
Phase 2	Logistic Regression + SMOTE	87.46%	78.36%	84.15%	81.15%
Phase 2	Random Forest (Tuned)	88.25%	84.09%	78.17%	81.02%
Phase 2	Gradient Boosting	88.70%	83.33%	80.99%	82.14%

What improved results

- Higher F1-score improved from 79.69% → 82.14%
- Better recall improved from 73.24% → 84.15% (LR+SMOTE)
- Better handling of class imbalance using SMOTE
- Ensemble models improved overall generalization
- Gradient Boosting gave the best balanced performance
- Better identification of dropout-risk students

System Integration

- Input ⟶ Processing ⟶ Model ⟶ Output



Demo / Results

```
============================================================
     SMART DROPOUT RISK CHECKER
============================================================

Enter Student Details:
Age: 19
Do you have a scholarship? (yes/no): no
Do you have any debt? (yes/no): no
Are your tuition fees up to date? (yes/no): yes
Semester 1: Subjects passed (0-6): 2
Semester 2: Subjects passed (0-6): 2

============================================================
                 RESULT
============================================================
Dropout Probability : 0.81
Risk Level          : 🔴 HIGH
Prediction          : Dropout

🔍 Key Reason:
 - Very low academic performance

💡 Suggested Action:
 - Focus on improving academics with tutoring and mentoring
```

Challenges Faced

- Understanding dataset features and their meaning
- Handling class imbalance in dropout data
- Selecting the most relevant features
- Choosing appropriate ML models
- Improving recall for dropout prediction

Future Work

Future Work

- Use larger and multi-college datasets
- Add more features like financial and behavioral data
- Try advanced models like XGBoost / LightGBM
- Deploy as a real-time dashboard or web app
- Use Explainable AI for better trust

Possible Improvements

- Better hyperparameter tuning
- Reduce false negatives
- Improve feature engineering
- Add real-time monitoring

Phase-2 Deliverables

- **Deliverable 1**: Cleaned dataset
- **Deliverable 2:** Jupyter Notebook
- **Deliverable 3:** Code Repository (GitHub)
  https://github.com/angelxavier905/AIML_ML-Lab/tree/main/ML_Mini
- **Deliverable 4:** System Integration Proof

References

- Kaggle Dataset: Predict Students' Dropout and Academic Success
- Villar & Andrade (2024) – Discover AI
- Marcolino et al. (2025) – Scientific Reports
- Cho et al. (2023) – Applied Sciences
- Kim et al. (2023) – Applied Sciences